

Anticonvulsant (Anti- epileptic) Agents

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Epilepsy

- The word 'Epilepsy' is derived from the Greek word 'epilepsia' meaning to be taken, seized or attacked.
- It is a very common disorder, characterized by seizures, which take various forms and result from episodic neuronal discharges
- **Seizure** – a sudden, forcible taking heading
 - a sudden attack of illness especially that of epilepsy
- **Epilepsy** - is a group of chronic seizure disorder having in common loss or disturbance of consciousness.
- **Seizure** - a brief episode of abnormal electrical activity in nerve cell of brain.

- Epileptic seizure can take two forms:

1. Generalized

- Grand mal (Tonic - clonic)
 - Complete loss consciousness
 - Muscular rigidity (Tonic)
 - General muscle spasm lasting for 2 to 5 minutes
 - The individual may stop breathing
 - Violent clonic convulsion (salivation, sweating)
- Absence seizure (petit mal)
 - Momentary loss of consciousness (free of convulsion)
 - The person stop doing something then continue

2. Localized (Partial)

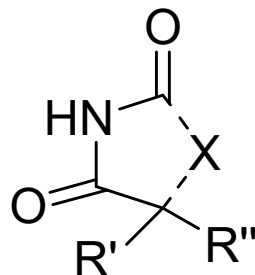
- Unilateral seizures involving one entire body side
 - Outburst of temper
 - Confused behavior

Anticonvulsant agents

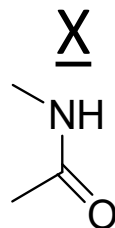
- Anticonvulsant acts selectively as depressant of convulsant activity in the brain
- The only effective way to control epilepsy is the use of drugs
- They all have common structure features
- **Classification**
 - Cyclic ureides and imides
 - Urea and
 - Miscellaneous

Cyclic ureides and imides

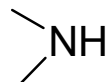
- **Classes**



- Barbiturate



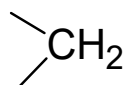
- Hydantoin



- Oxazolidindiones

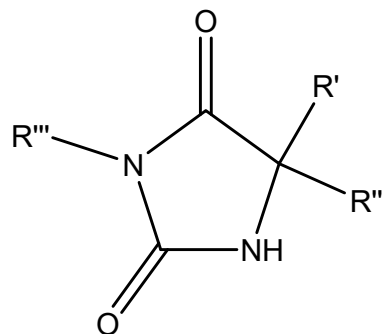


- succinimides



- If both are lower alkyl, the tendency is to be active against petit mal
- If one is an aryl group, activity tends to be directed to grand mal epilepsy
- It is believed the imide hydrogen atom is involved in a strong hydrogen bond at the receptor
- The strength of activity depends on the substituents

Hydantoins



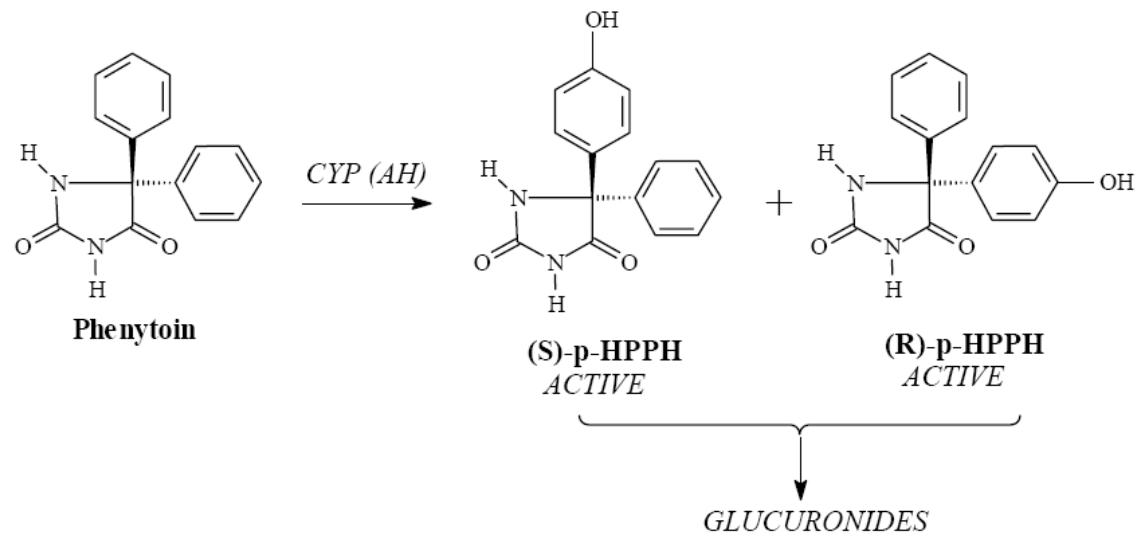
Name	R'	R''	R'''
Phenytoin	C ₆ H ₅	C ₆ H ₅	H
Ethotoin	C ₂ H ₅	C ₆ H ₅	H
Mephentyoin	C ₂ H ₅	C ₆ H ₅	CH ₃

- Phenytoin and the other hydantoins are “ring-contracted” analogues of the barbiturates (one carbonyl removed).
- These compounds have physicochemical properties similar to the barbiturates, including acidity when $R = H$ as in phenytoin (an acidic imide with pK_a 8.3).

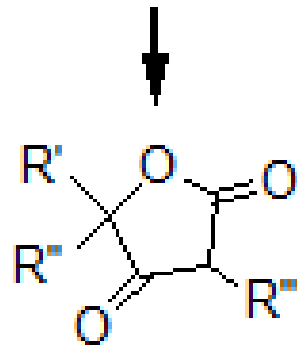
- **MOA: depression of the sodium action potential, "filtering out" sustained high frequency neuronal discharges and synaptic activity.**
- **This may be related to a voltage dependent blockade of membrane sodium channels responsible for the action potential.**

- **Phenytoin** is also a weak **antiarrhythmic** and these actions are also mediated through effects on sodium channels, in this case, in Purkinje fibers.
- At least one phenyl group is important for activity
- Phenytoin (a prochiral compound) is metabolized in the liver mainly to enantiomeric phenols similar to aromatic barbiturates.

- The oxidative metabolism of phenytoin is capacity-limited and shows saturability.
- These metabolites may be conjugated as glucurionides and excreted in the urine by tubular secretion.



Oxazolidinediones



Name

R'

R''

R'''

Trimethadione

CH₃

CH₃

CH₃

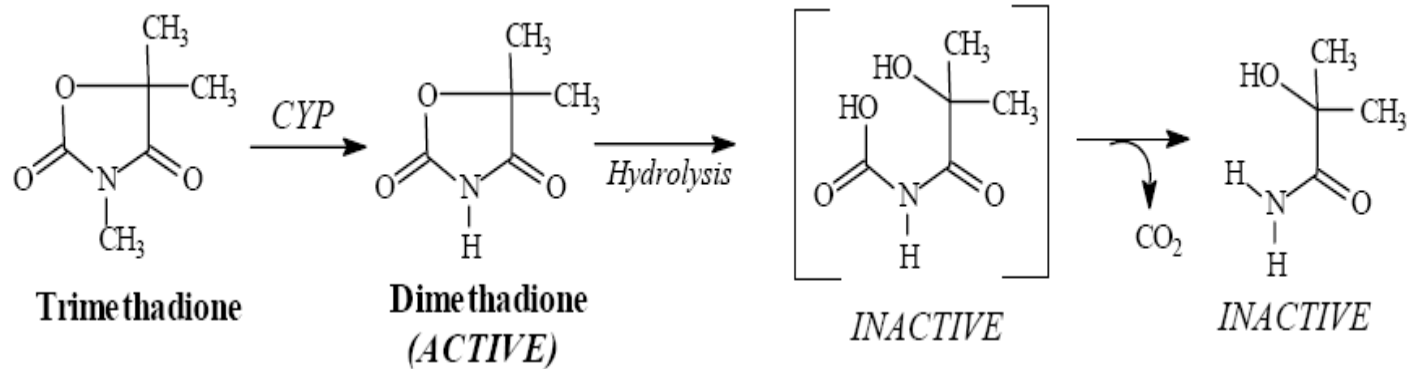
Paramethadione

C₂H₅

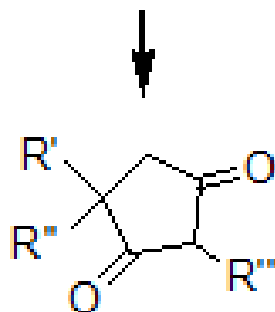
CH₃

CH₃

- Trimethadione and the other oxazolidinediones are “ring-contracted” analogues of the barbiturates (one carbonyl removed) or can be viewed as isosteres of the hydantoins (one ring N replaced with O).
- These compounds have physicochemical properties similar to the barbiturates, except they are not acidic (no imide).



Succinimides



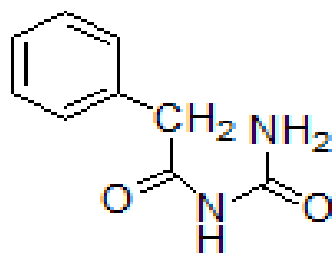
Name	R'	R''	R'''
Ethiosuccinimide	C ₂ H ₅	CH ₃	H
Methosuccinimide	C ₆ H ₅	CH ₃	CH ₃
Phensuccinimide	C ₆ H ₅	H	CH ₃

- Ethosuximide and the other succinimides are “ringcontracted” analogues of the barbiturates (one carbonyl removed) or can be viewed as isosteres of the hydantoins (one ring N replaced with CH₂).
- These compounds have physicochemical properties similar to the barbiturates, except only ethosuximide has an imide N-H and is acidic.

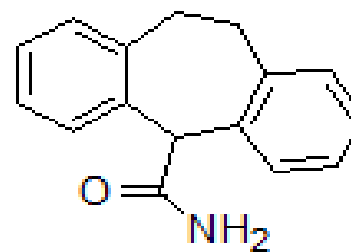
- The antiepileptic (petit mal) actions of the succinimides may be related to synaptic inhibition brought about by the GABA mediated chloride conductants

✓

Ureas

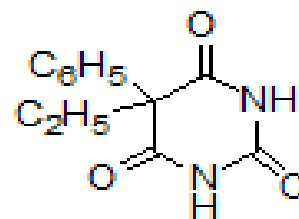
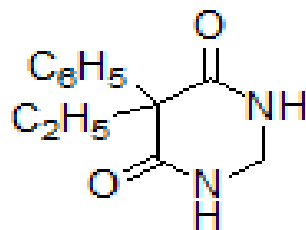


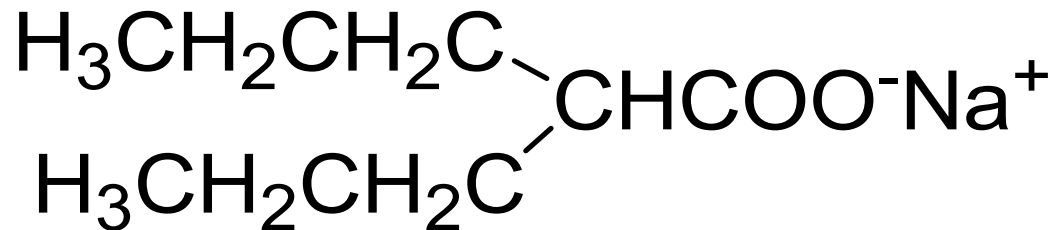
Phenacetamide



Carbamazepine

c) Miscellaneous





Sodium Valproic acid

- Valproic acid blocks GABA transaminase
- Elevating GABA level and thus alleviating seizure
- GABA transaminase is responsible for the destruction of GABA

Primidone :Is metabolized to **phenobarbitone** and it acts through its metabolite